

B.Tech. III Semester (EE/ICE)

END-SEMESTER EXAMINATION, NOV.-DEC. 2024

Course Code: ICECC08/ EEECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 3:00 Hours

Max. Marks: 50

Note: Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Attempt any TWO parts (a) Design a combinational logic circuit that detects the presence of any of the six illegal code groups in the Excess-3 code by providing a logic-1 output. Implement it by using NOR gates only. (b) Design a circuit that multiplies two numbers of two bits each and implement using NAND gates only. (c) Design a circuit that generates the 2's complement of the 4 bit number and implement using gates.	5 5 5	CO1 CO1 CO1
2	Attempt any TWO parts (a) Using the Quine-McClusky (Tabulation) method minimize the function given below having don't care terms $f(w, x, y, z) = \prod_M (1, 3, 9, 11, 12, 13) + \text{don't care}(2, 10, 14, 15)$ (b) Design a circuit that converts BCD code to 2421 code and implement it by using a suitable size Decoder and OR/NOR gates. (c) Implement the following functions using a cost efficient PLA assuming both true and complementary form are available at the output. Draw PLA table. $F1(A, B, C, D) = \sum_m (0, 1, 2, 3, 6, 7)$ $F2(A, B, C, D) = \sum_m (0, 1, 2, 3, 8, 9)$ $F3(A, B, C, D) = \sum_m (0, 1, 6, 7, 14, 15)$	5 5 5	CO2 CO2 CO2

3	<u>Attempt any TWO parts</u> (a) Using a positive edge triggered JK Flip-Flop design a synchronous mod-5 counter whose counting sequence is $110 \rightarrow 101 \rightarrow 111 \rightarrow 001 \rightarrow 011 \rightarrow 110$	5	CO3
	(b) Convert the positive edge triggered T flip flop into positive edge triggered S-R flip flop.	5	CO3
	(c) Differentiate between Ring counter and Johnson counter. Give their applications.	5	CO3
4	<u>Attempt any TWO parts</u> (a) Write VHDL code for the circuit that adds two numbers of 4 bit each.	5	CO4
	(b) Write a VHDL code for a 4 bit UP counter, having clear and preset inputs, and implemented using positive edge triggered J-K flip flop..	5	CO4
	(c) Write a VHDL code to implement 3X8 Decoder.	5	CO4
5	<u>Attempt any TWO parts</u> (a) Draw and explain in detail the NAND and NOR gates fabricated using CMOS logic family.	5	CO1
	(b) Explain in detail ROM and PLA.	5	CO5
	(c) What is meant by PLD? Draw and explain the basic architecture of CPLD and FPGA.	5	CO5

Paul